10.11 Three Main components of robots

1. Mechanical Components
2. Electrical components
3. Computer programming code

BASIC PRINCIPLE OF ROBOTICS

10.12 Asimov Laws of robotics

1. A robot must not injure a human being or, through inaction, allow a human being to come harm.
2. A robot must obey the orders given it by human being except where such orders would conflict with first law.
3. A robot must protect its own existence as long as such protection does not conflict with the first law or second law.

Mr. Isaac Asimov
(1920 - 1992)



Isaac Asimov was an American writer and professor of biochemistry at Boston University. He was known for his works of science fiction and popular science. He was a prolific writer who wrote (or) edited more than 500 books and an estimated 90,000 letters and postcards.

10.13 Need of Robotics

- To improve the quality of products
- To reduce the preparation time
- To reduce the rejection rate and waste
- Higher flexibility of product type and variation
- Skilled labour shortage
- Rising cost
- Pressure to increase production rates to complete market.

Functions of Robotics

It can be classified into three areas

1. "Sensing" the environment by external sensors.

   Example: Vision, Voice, Touch, Proximity, and so on
2. "Decision making" based on the information received from the sensor
3. "Performing" the task decided.

10.14 Types of Robots

1. Based on structure
2. Based on application
   a. Assembly robot
   b. Welding robot
   c. Heavy duty robot
   d. Military robot
   e. Hospital robot
   f. Cleaning robot
   g. Agricultural robot
   h. Medical robot
   i. House hold robot
   j. Nano robot
3. Based on coordinate system,
   a. Base on the reference point
      i. Base coordinate system
      It is located at the base of the robot
      ii. The work piece coordinate system
      It is related to the work piece and is often the best one for programming the robot.

XII BME_English version CHAPTER 10.indd 173 1/8/2020 1:37:58 PM